
29510: Project 5

Sammy Liu, Allen Wang

Overview

01 Initial Testing: Constructors, Improvers, Guiders

02 Basic HGS

03 Modified HGS

04 Benchmarks

Three building blocks

CONSTRUCTOR

find an initial solution

IMPROVER

find the next step

GUIDER

shape the search

Constructors

[C1]

farthest_nearest

Seed at customer farthest from depot; greedily extend by nearest.

[C2]

regret_bidding

Insert customer with largest gap between best & 2nd-best insertion cost.

[C3]

sweep

Partition by polar angle from depot; route each cluster.

Improvers

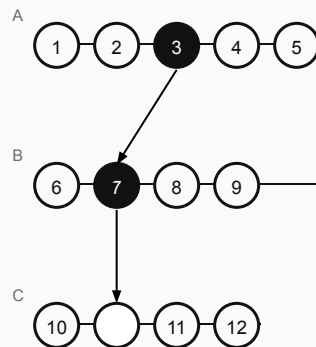
[I1] destroy_repair

Remove a chunk; reinsert greedily.



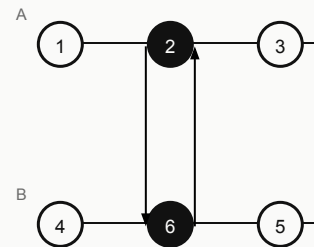
[I2] ejection_chains

Push out → propagate displacement down a chain.



[I3] relocate_swap

Move or swap single customers between routes.



Guiders

[G1]

tabu

Short-term memory of recent moves; reversals are forbidden until they expire.

[G2]

granular

Restrict the neighborhood to closest customers only - long distances are pruned.

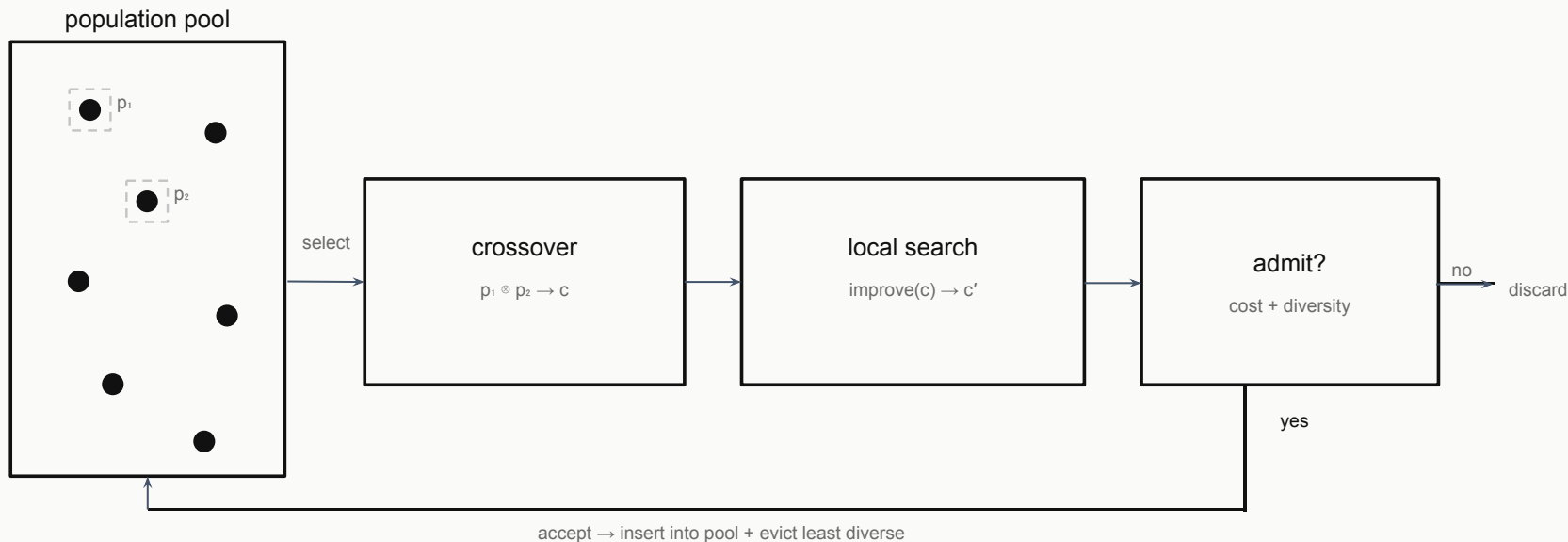
Constructor screening - wins
(best objective on each of 16 instances)

	wins
regret_bidding	6
farthest_nearest	5
sweep	5

Avg % improvement over constructor baseline — improver × guidance
(10 s runs, all instances, seed 0)

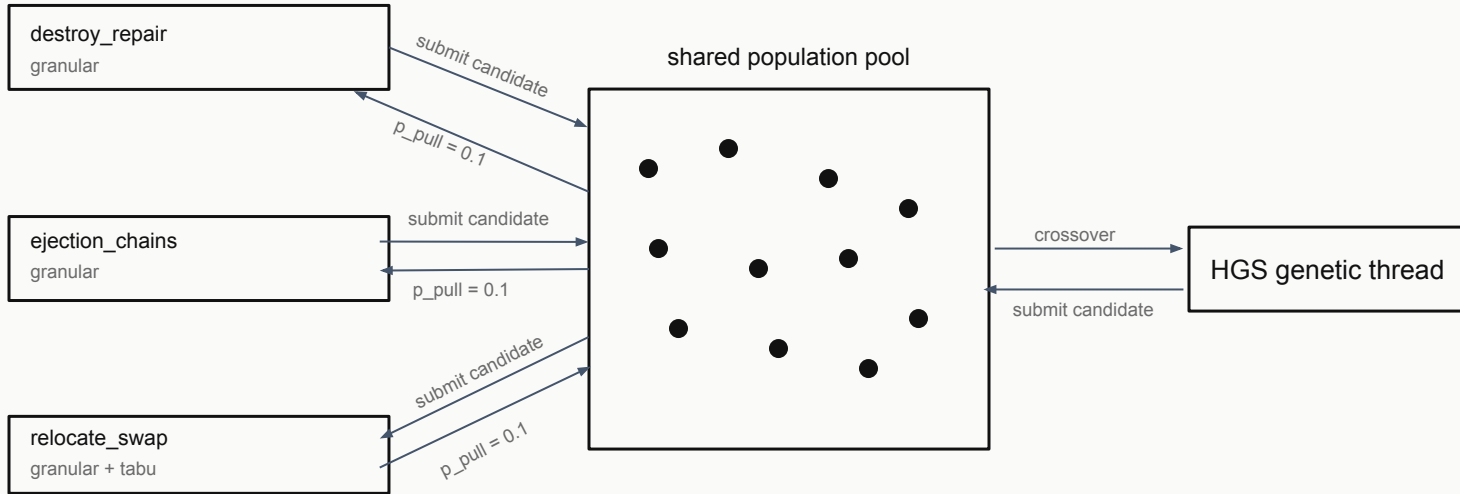
	none	granular	granular + tabu
destroy_repair	6.75%	9.03%	8.32%
ejection_chains	3.83%	4.91%	4.66%
relocate_swap	4.52%	3.64%	3.48%

Standard Hybrid Genetic Search (HGS)



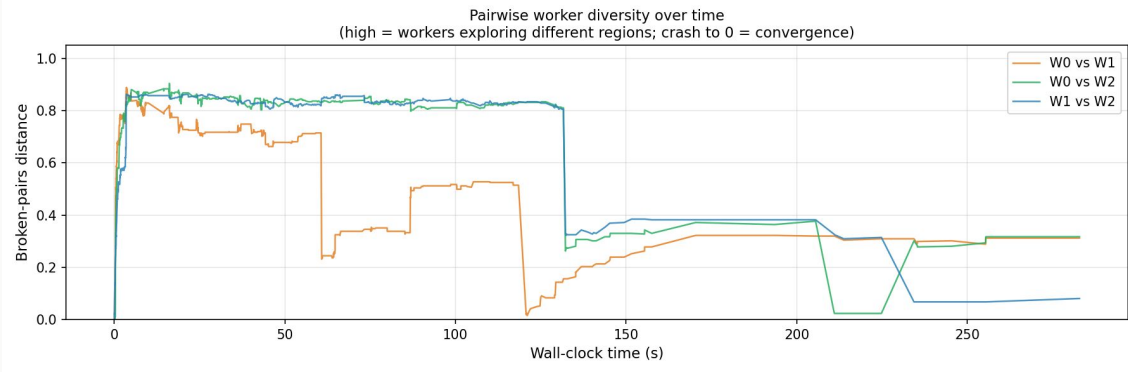
MultiWorker HGS

3 ALNS workers + 1 HGS thread, shared pool

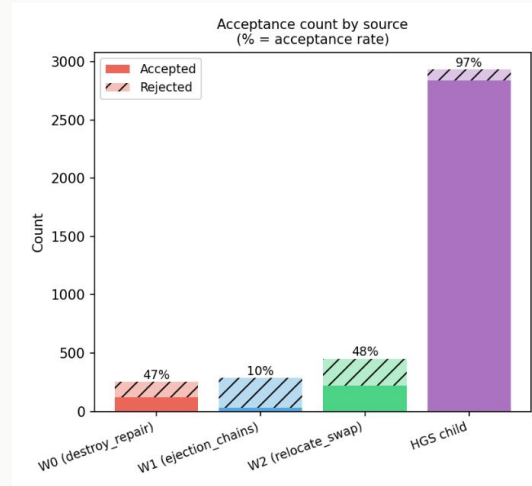


Workers stay diverse, get accepted

pairwise broken-pairs distance x wall-clock - workers stay mostly separated at ~0.8



acceptance rate by source



BENCHMARK: Standard HGS vs MultiWorker HGS

instance	baseline obj HGS standalone	bidirectional obj 3-worker MT	% diff neg = MT better
16_5_1	334.96	334.96	—
21_4_1	358.40	358.40	—
30_4_1	505.01	505.01	—
41_14_1	861.79	861.79	—
45_4_1	723.54	723.54	—
51_5_1	524.61	524.61	—
76_8_2	740.66	740.66	—
101_11_2	1,940.61	1,939.90	-0.04%
101_8_1	827.39	826.14	-0.15%
121_7_1	1,042.12	1,042.12	—
135_7_1	1,162.96	1,162.96	—
151_15_1	3,083.92	3,056.41	-0.89%
200_16_2	1,487.98	1,349.25	-9.32%
241_22_1	745.73	712.85	-4.41%
262_25_1	5,955.46	5,563.43	-6.58%
386_47_1	25,897.20	24,929.19	-3.74%

Thanks!

Time spent: 30 hours
